

NEPALI LIFESTYLE-BASED DIET AND FITNESS ADVISORY SYSTEM



PRESENTED BY:

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INTRODUCTION

- This project is an AI-based Nepali Diet Advisory System that identifies Nepali food from images and provides its nutritional information.
- Based on the nutritional information and user profile, system provides diet and fitness guidance.

PROBLEM STATEMENT

- Existing systems often provide generic recommendations rather than user-specific advice.
- Limited consideration of Nepali food items, eating habits, and lifestyle patterns.

OBJECTIVES

- To build a personalized application that tracks daily user nutrition intake and provides personalized, easy-to-understand health recommendations.
- To generate personalized diet and fitness recommendations based on the user's profile, food intake, and nutritional requirements.

LITERATURE REVIEW

No.	Reference	Key Findings	Limitation
1.	Tsolakidis et al. (2024)	Showed that AI-based personalized nutrition is moving beyond basic calorie counting toward adaptive recommendations.	Generic food databases may not accurately support regional diets such as Nepali cuisine.
2.	Jiang et al. (2026) – DietGlance	Used smart glasses and knowledge bases to provide real-time dietary feedback.	It requires specialized wearable device, making it costly and less accessible.
3.	Manandhar et al. (2025) – SmartCal	Demonstrated the feasibility of automatically detecting common Nepali foods using computer vision and Mask R-CNN.	Mask R-CNN is computationally demanding and can introduce high latency on mobile/edge devices.

TOOLS AND TECHNOLOGIES USED

- YOLOv11 : For household Nepali plate food item detection and segmentation.
- ChormaDB: As vector database to store food and nutrition data embedding.
- Kaggle, Roboflow and iscraper: For data collection and scrapping.
- React: For frontend.
- Django: For backend.
- FastAPI: For microservies.

SYSTEM BLOCK DIAGRAM

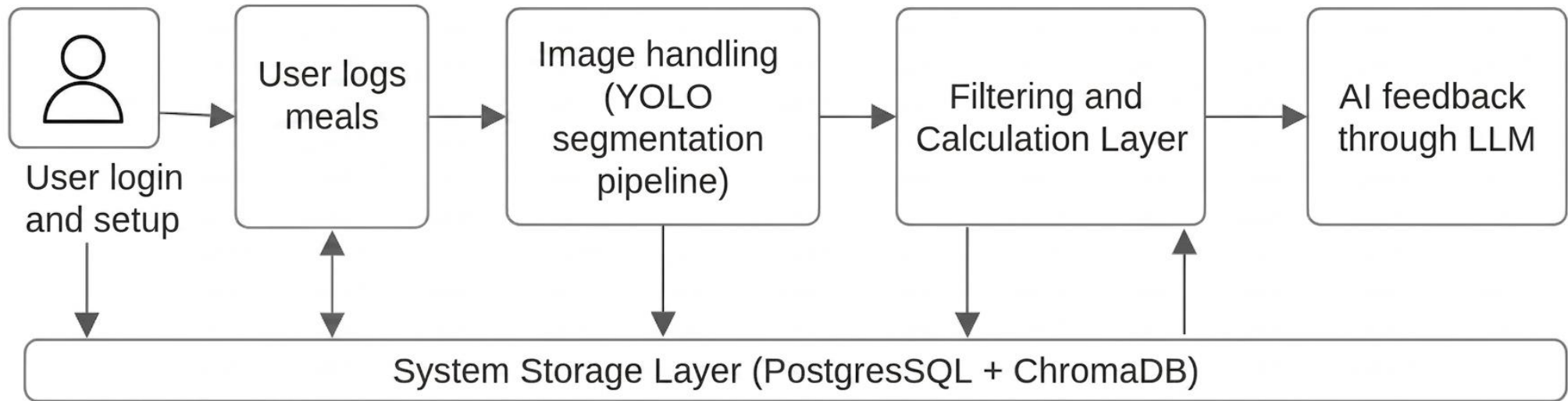
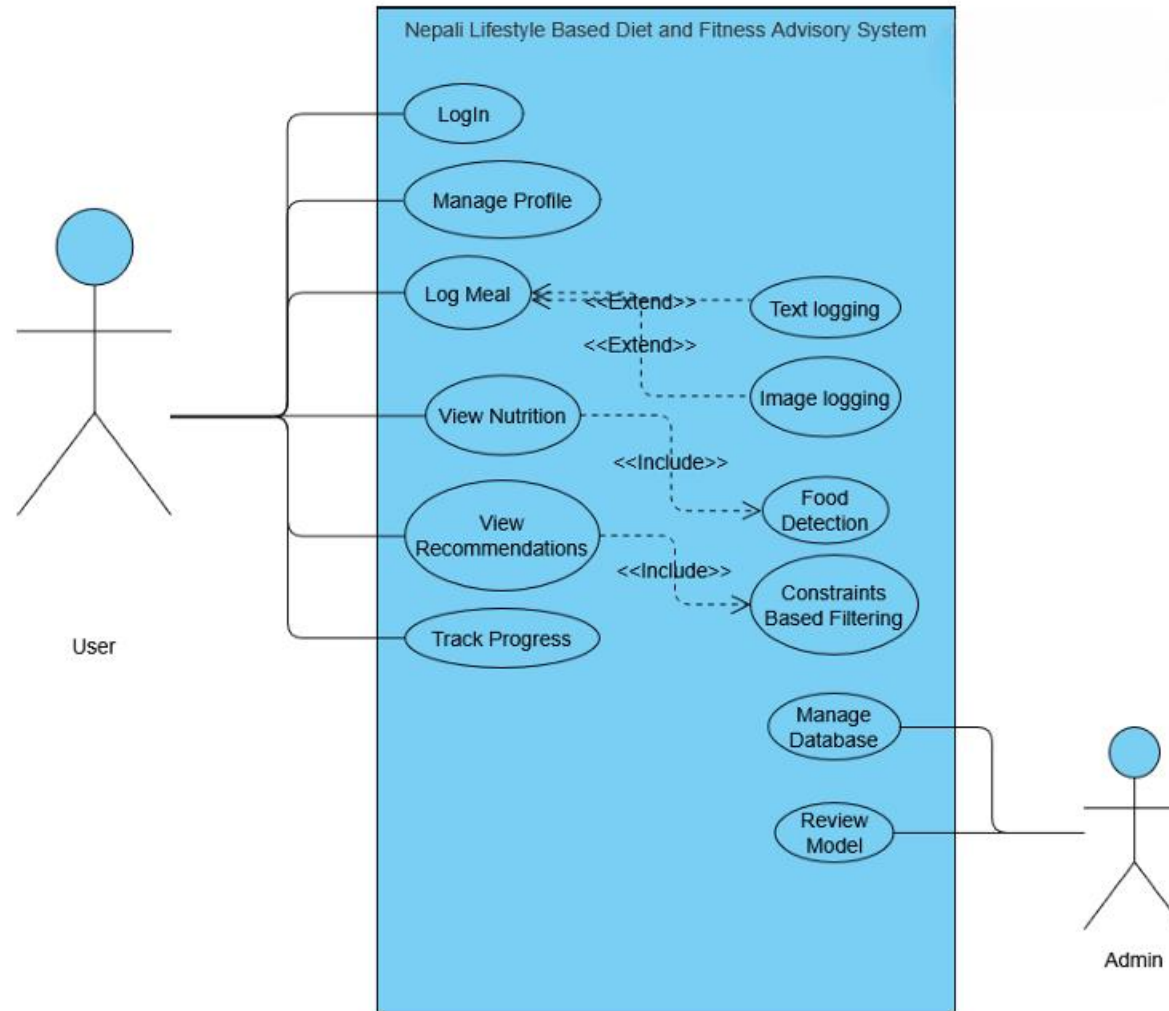
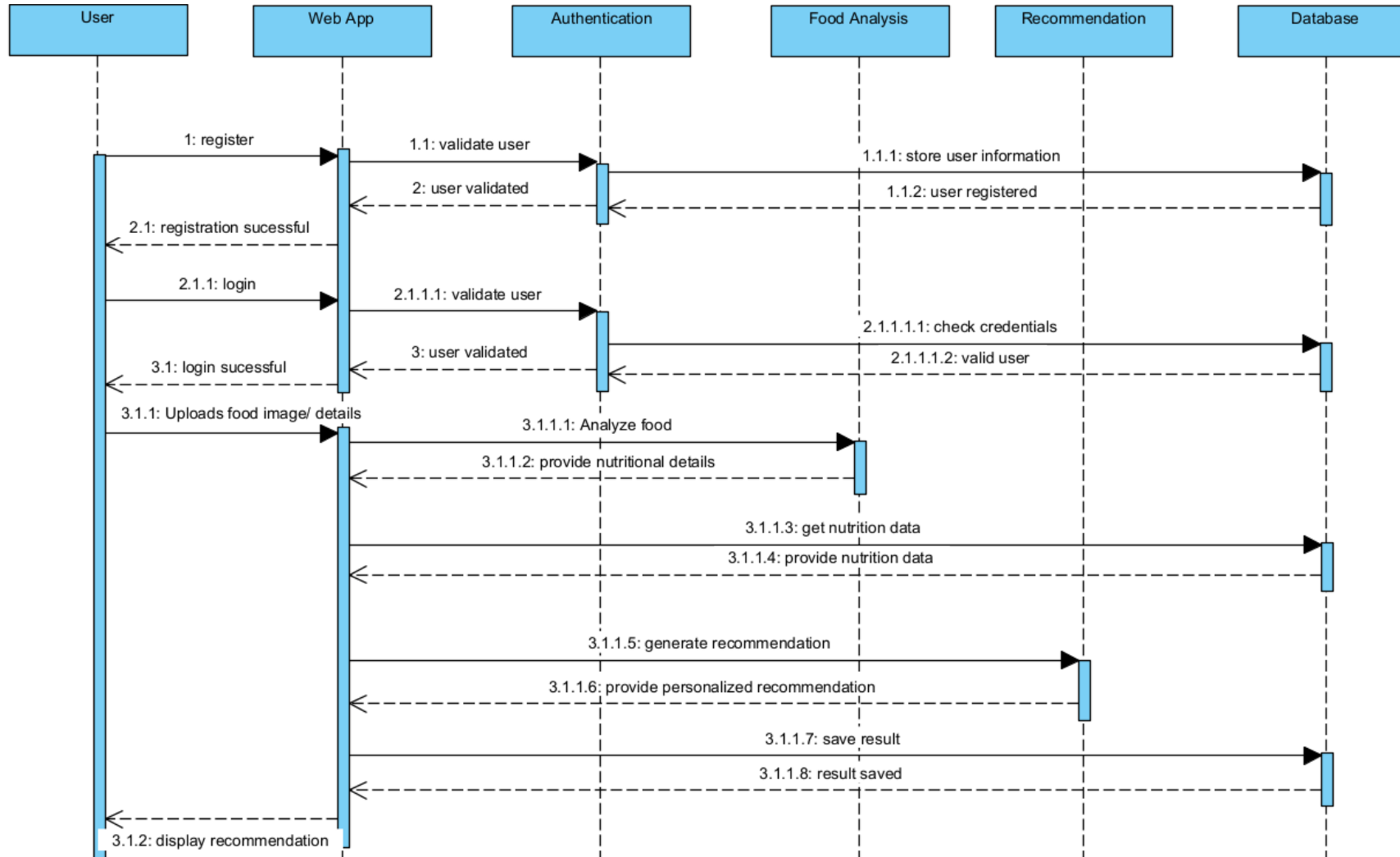


Fig: Proposed System Architecture and Data Processing Workflow

USECASE DIAGRAM



SEQUENCE DIAGRAM



METHODOLOGY

- User creates a personal profile with physical metrics, health goals, and dietary constraints.
- Meal inputs are logged using either plate photo upload or informal text descriptions.
- Text inputs is parsed and goes through a similarity search with our database.
- Plate images undergo YOLO segmentation to detect food items and estimate portion weights.
- A local LLM processes total calculated nutrition against user profile data to output tailored dietary advice.

PHOTO WORKFLOW

- **Image Capture:** User uploads a plate photo via React UI, routed through Django to the FastAPI AI service.
- **Segmentation:** YOLO model (best.pt) detects individual food items and extracts area masks.
- **Volume Estimation:** Volume and weight is estimated using our custom logic.
- **Nutrient Match:** Correlates estimated grams with the database to compute total calories, protein, carbs, and fats.
- **Advisory Generation**

TEXT WORKFLOW

- **Input Parsing:** User types meal logs in English or Romanized Nepali.
- **Entity Extraction:** Natural language parser identifies food names and quantity units (e.g., cups, plates, grams).
- **Vector Matching:** Semantic search matches extracted names to exact Nepali food database records.
- **Nutrition Scaling:** Calculates nutritional totals based on specified or default portion sizes.
- **Response Delivery:** Saves meal history to DB and displays calculated macro breakdown to the frontend.

DATA FLOW DIAGRAM

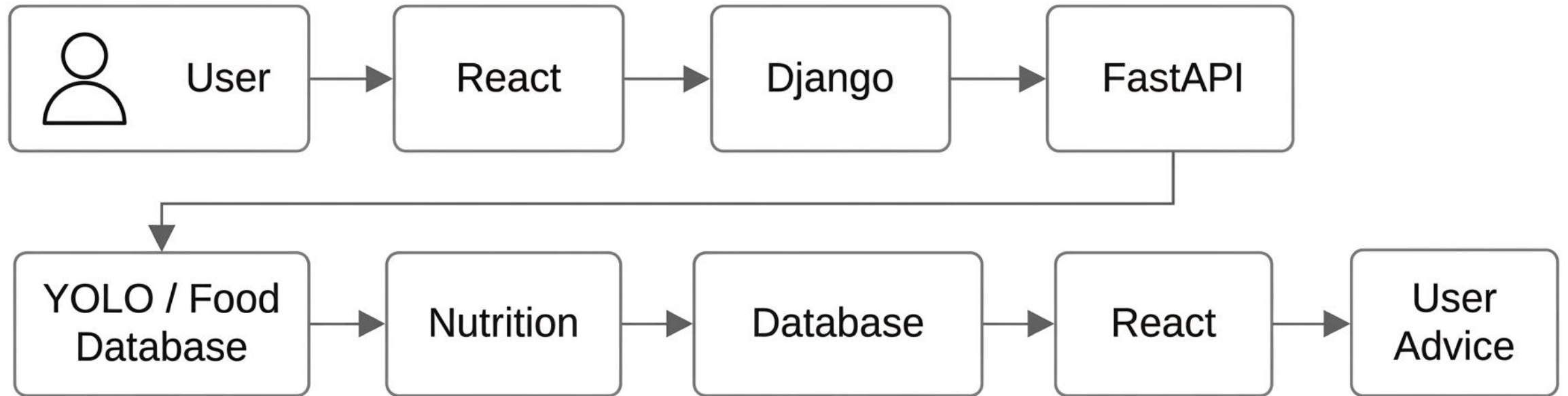


Fig: Data flow

GAPS AND CHALLENGES

- Limited availability of reliable and structured nutritional data for diverse Nepali foods.
- NutriNepal is a AI based system and can make mistakes. It is not intended to replace professional doctors.

RESULT AND CONCLUSION

- Developed an AI-powered system for identifying Nepali food items from images and meal information.
- Provides nutritional analysis and personalized diet recommendations based on individual user profiles.
- Supports different local and colloquial food names through semantic matching.

FUTURE ENHANCEMENTS

- Broaden regional food data and local nutritional details to boost analysis accuracy.
- Improve food matching, portion estimation, personalized recommendations, and Nepali-language support.
- Increase the options for social and health based constraints.

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THANK YOU